

D625

Principles of Epidemiology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A town of 20,000 had 400 people with diabetes on January 1. During the year, 100 new cases were diagnosed. The town's clinic recently extended its hours. What was the one-year incidence proportion?

- A. 5.0 per 1,000.
- B. About 5.1 per 1,000.
- C. 20 per 1,000.
- D. 25 per 1,000.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A chronic condition has an incidence of 4 per 1,000 people per year, and people live with it an average of 5 years. It was first described by researchers in the 1980s. Assuming steady conditions, what is its approximate prevalence?

- A. 0.8 per 1,000.
- B. 9 per 1,000.
- C. 4 per 1,000.
- D. 20 per 1,000.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A condition has a prevalence of 30 per 1,000 and an incidence of 10 per 1,000 per year. What is the approximate average duration?

- A. 0.33 years.
- B. 3 years.
- C. 20 years.
- D. 300 years.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A 10-year cohort run from a university hospital follows 2,000 smokers and 4,000 nonsmokers. Sixty smokers and 20 nonsmokers develop the disease. What is the relative risk for smokers?

- A. 25 per 1,000
- B. 6.0
- C. 3.0
- D. 0.17

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

An exposure has a relative risk of 4. Risk in the unexposed is 2 per 1,000. What is the risk in the exposed?

- A. 6 per 1,000.
- B. 0.5 per 1,000.
- C. 8 per 1,000.
- D. 4 per 1,000.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Researchers recruit 200 patients with a rare liver disease and 400 controls from the same clinics, which serve three counties. Of the cases, 120 report past solvent exposure; of the controls, 100 do. What is the odds ratio?

- A. 2.4
- B. 0.22
- C. 4.5
- D. 1.2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A state wants to study whether a pesticide is linked to a childhood cancer that affects about 1 in 50,000 children. The state has a well-staffed cancer registry. Which design is most efficient?

- A. A cohort is always the strongest design.
- B. Randomized trial assigning pesticide exposure.
- C. Case-control study.
- D. Prospective cohort study.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A study finds heavy coffee drinkers have higher lung cancer rates. Coffee drinkers in the sample are also much more likely to smoke. Most participants were recruited at a regional fair. What best explains the finding?

- A. Smoking is a mediator between coffee and cancer.
- B. Coffee causes lung cancer.
- C. Selection bias from recruiting at a fair.
- D. Confounding by smoking; adjust or stratify by smoking.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Exercise lowers blood pressure, which lowers stroke risk. In a study of exercise and stroke, which variable is a mediator rather than a confounder?

- A. Blood pressure.
- B. Age.
- C. Income.
- D. No variable can be a mediator.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A one-time survey of 5,000 adults, funded by a state grant, finds depression is strongly linked to unemployment (odds ratio 3.2). Which Bradford Hill criterion can this study not establish?

- A.** Temporality.
- B.** Strength of association.
- C.** A strong association proves cause.
- D.** Consistency.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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11.

A test with 90% sensitivity and 95% specificity is used in a clinic of 1,000 patients where 10% have the disease. The test kit won an innovation award. What is the positive predictive value?

- A.** 90%.
- B.** 95%.
- C.** About 99%.
- D.** About 67%.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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12. [TRANSFER — reversed stem / different variable]

The same test (90% sensitivity, 95% specificity) is used in a community of 10,000 where only 1% have the disease. What is the positive predictive value?

- A.** About 15%.
- B.** 1%.
- C.** About 67%.
- D.** 90%.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A town of 20,000 had 400 people with diabetes on January 1. During the year, 100 new cases were diagnosed. ~~The town's clinic recently extended its hours.~~ What was the one-year incidence proportion?

Target word: incidence **Decides:** New cases ÷ people at risk (20,000 – 400).

A	Dropped element	Leaves existing cases in the at-risk denominator.
B	Correct	$100 \div 19,600 = 5.1$ per 1,000.
C	Wrong stem number	Starting prevalence (400 ÷ 20,000).
D	Confusion set	Year-end prevalence (500 ÷ 20,000).

2. Answer D Concept 2

A chronic condition has an incidence of 4 per 1,000 people per year, and people live with it an average of 5 years. ~~It was first described by researchers in the 1980s.~~ Assuming steady conditions, what is its approximate prevalence?

Target word: prevalence **Decides:** $P \approx I \times D = 4 \times 5$.

A	Inverted operation	Divides incidence by duration.
B	Confusion set	Adds incidence and duration.
C	Dropped element	Ignores duration.
D	Correct	4 per 1,000 per year × 5 years.

3. Answer B Concept 2 • transfer

A condition has a prevalence of 30 per 1,000 and an incidence of 10 per 1,000 per year. What is the approximate average duration?

Target word: solve for duration **Decides:** $D \approx P \div I$.

A	Confusion set	Divides incidence by prevalence.
B	Correct	$30 \div 10 = 3$.
C	Adjacent category	Subtracts incidence from prevalence.
D	Inverted operation	Multiplies instead of dividing.

4. Answer B Concept 3

A 10-year cohort ~~run from a university hospital~~ follows 2,000 smokers and 4,000 nonsmokers. Sixty smokers and 20 nonsmokers develop the disease. What is the relative risk for smokers?

Target word: relative risk **Decides:** $(60/2,000) \div (20/4,000)$.

A	Adjacent category	That is the risk difference.
B	Correct	$30 \text{ per } 1,000 \div 5 \text{ per } 1,000 = 6$.
C	Dropped element	Divides counts (60 ÷ 20) without group sizes.
D	Inverted operation	Unexposed over exposed.

5. Answer C Concept 3 • transfer

An exposure has a relative risk of 4. Risk in the unexposed is 2 per 1,000. What is the risk in the exposed?

Target word: solve for exposed risk **Decides:** $RR \times \text{unexposed risk}$.

A	Confusion set	Adds RR to the risk.
B	Inverted operation	Divides by RR.
C	Correct	$4 \times 2 = 8$.
D	Wrong stem number	The RR, not a risk.

6. Answer C Concept 4

Researchers recruit 200 patients with a rare liver disease and 400 controls from the same clinics, ~~which serve three counties~~. Of the cases, 120 report past solvent exposure; of the controls, 100 do. What is the odds ratio?

Target word: odds ratio **Decides:** $(120 \times 300) \div (80 \times 100)$.

A	Adjacent category	A ratio of proportions ($0.60 \div 0.25$), not odds.
B	Inverted operation	The reciprocal of the correct OR.
C	Correct	$36,000 \div 8,000 = 4.5$.
D	Dropped element	Divides exposed counts only ($120 \div 100$).

7. Answer C Concept 5

A state wants to study whether a pesticide is linked to a childhood cancer that affects about 1 in 50,000 children. ~~The state has a well-staffed cancer registry.~~ Which design is most efficient?

Target word: design choice **Decides:** The disease is rare.

A	Absolute language	Design fit depends on rarity and ethics.
B	Confusion set	Unethical to assign a possible harm.
C	Correct	Starts from the rare cases.
D	Adjacent category	Would need a huge cohort for enough cases.

8. Answer D Concept 6

A study finds heavy coffee drinkers have higher lung cancer rates. Coffee drinkers in the sample are also much more likely to smoke. ~~Most participants were recruited at a regional fair.~~ What best explains the finding?

Target word: name the problem **Decides:** Smoking is tied to coffee and independently causes lung cancer.

A	Inverted operation	Coffee does not cause smoking.
B	Dropped element	Ignores the smoking link.
C	Camouflage	The venue does not explain the coffee-smoking link.
D	Correct	Smoking meets both conditions for a confounder.

9. Answer A Concept 6 • transfer

Exercise lowers blood pressure, which lowers stroke risk. In a study of exercise and stroke, which variable is a mediator rather than a confounder?

Target word: identify mediator **Decides:** On the causal pathway.

A	Correct	It lies between exposure and outcome.
B	Inverted operation	Affects both exercise and stroke: a confounder.
C	Adjacent category	Can affect both; a possible confounder.
D	Absolute language	Mediators are common.

10. Answer A Concept 7

A one-time survey of 5,000 adults, ~~funded by a state grant~~, finds depression is strongly linked to unemployment (odds ratio 3.2). Which Bradford Hill criterion can this study not establish?

Target word: unmet criterion **Decides:** Both measured at once; order unknown.

A	Correct	A single survey cannot show which came first.
B	Wrong stem number	OR 3.2 does speak to strength.
C	Absolute language	Strength alone is not proof.
D	Adjacent category	Needs other studies, not this design flaw.

11. Answer D Concept 8

A test with 90% sensitivity and 95% specificity is used in a clinic of 1,000 patients where 10% have the disease. ~~The test kit won an innovation award~~. What is the positive predictive value?

Target word: PPV **Decides:** TP = 90; FP = 45 (5% of 900).

A	Confusion set	That is sensitivity.
B	Adjacent category	That is specificity.
C	Inverted operation	Negative predictive value: $855 \div 865$.
D	Correct	$90 \div (90 + 45) = 66.7\%$.

12. Answer A Concept 8 • transfer

The same test (90% sensitivity, 95% specificity) is used in a community of 10,000 where only 1% have the disease. What is the positive predictive value?

Target word: PPV at low prevalence **Decides:** TP = 90; FP = 495.

A	Correct	$90 \div 585 = 15.4\%$.
B	Wrong stem number	That is prevalence.
C	Dropped element	Ignores the drop in prevalence.
D	Confusion set	Sensitivity does not change with prevalence; PPV does.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Incidence vs. prevalence	Incidence counts new cases among people at risk over a period; prevalence counts all existing cases at a point or period.	Leaving existing cases in the at-risk denominator, or counting old cases as new.	Existing cases at the start plus new cases during the period.
2	Prevalence $\approx$ incidence $\times$ duration	For a stable condition, prevalence is about incidence times average duration, so longer survival without cure raises prevalence.	Thinking prevalence rises only when new cases rise.	An incidence rate and an average disease duration.
3	Relative risk from a cohort study	Relative risk = risk (incidence) in the exposed $\div$ risk in the unexposed.	Dividing raw case counts without using each group's size.	Exposed and unexposed groups followed over time, with group sizes.
4	Odds ratio from a case-control study	OR = (exposed cases $\times$ unexposed controls) $\div$ (unexposed cases $\times$ exposed controls).	Computing a risk ratio from case-control data.	Groups chosen by disease status, then asked about past exposure.
5	Choosing a study design	Case-control studies suit rare diseases; cohort studies suit rare exposures and measure incidence; trials are used only when assigning the exposure is ethical.	Following a large cohort for a very rare disease, or proposing a trial of a harmful exposure.	Whether the disease or the exposure is rare, and whether exposure could harm.
6	Confounding	A confounder is linked to both exposure and outcome and is not on the causal path; it distorts the association until controlled by stratifying, matching, or adjusting.	Treating a step on the causal pathway (a mediator) as a confounder.	A third factor that travels with the exposure and independently causes the outcome.
7	Bradford Hill criteria: temporality	Of the causal criteria, only temporality (exposure before outcome) is essential.	Treating a strong association as proof of cause without knowing which came first.	Exposure and outcome measured at the same time.
8	Screening accuracy and predictive value	Sensitivity and specificity are properties of the test; positive predictive value = true positives $\div$ all positives and falls as prevalence falls.	Equating sensitivity with the chance that a positive result is true.	Prevalence, sensitivity, specificity, and a question about a positive result.

Course scope (public description): Principles of Epidemiology offers an in-depth exploration of foundational principles and methodologies crucial for discerning and interpreting health and disease patterns in populations. Learners engage with core epidemiological concepts, encompassing measures of disease frequency, study designs, and principles of causation. Emphasizing practical applications, the course equips learners with analytical skills and theoretical foundations essential for conducting epidemiological research and contributing to evidence-based public health practices. Delving into a diverse range of topics, participants cultivate a robust understanding of epidemiological principles and their real-world applications, facilitating a seamless integration of knowledge into public health practices. There are no prerequisites for this course.